Four Failed Rescue Attempts for the Homology Cliff: Mahalanobis, Fisher-Rao, Stratified Cascade, and Mapper-Biased Panel Augmentation

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April 12, 2026 (v1.1; erratum 2026-06-15)

Abstract

The homology cliff in frozen ESM-2 [1] biosecurity retrieval (Paper 1) admits many natural rescue hypotheses. We pre-register and reject four. (i) Ledoit-Wolf [2] Mahalanobis whitening reduces the close-distant F_1 gap only by collapsing close-stratum F_1 ($0.866 \to 0.456$ at t30 R=1000 k=25); distant F_1 worsens ($0.120 \to 0.087$). (ii) Fisher-Rao [3] within-class whitening underperforms Mahalanobis in 17 of 18 pre-registered cells, with penalty up to $+0.285$ F_1. (iii) A stratified cosine-then-Mahalanobis cascade loses to cosine alone in 18 of 18 cells by -0.046 to -0.310 pooled F_1. (iv) A Mapper [4]-biased panel drawn from positive-enriched topological regions produces distant-F_1 rescue -0.0080, 95% CI $[-0.035, +0.018]$, indistinguishable from zero. Cross-family Pfam partition (Paper 5) independently finds all 20 evaluable distant false alarms are cross-family (Wilson 95% CI on the within-family rate $[0\%, 17\%]$), substantially weakening the assumption that panel expansion within failing families could correct. Taken with Paper 1's positive learned-projection result, the dichotomy is sharp: post-hoc distance substitution and panel sampling tricks fail at pre-registered H1, while embedding-space rotation succeeds. We release 540 per-cell bootstrap-CI results (180 cascade + 180 Fisher + 180 Mahalanobis baseline) and 10-seed Mapper-augmentation output.

Erratum (2026-06-15). Attempt 4 (Mapper-biased panel augmentation): the biased-panel positive pool is now deduplicated. The 30%-overlapping Mapper cover caused boundary proteins to recur across node member lists, and `rng.choice(replace=False)` removes positions, not values, so the biased arm held fewer than 500 unique positives. After the fix the rescue is -0.0080 (was $+0.0018$), 95% CI $[-0.035, +0.018]$; the CI still spans zero, H1 remains rejected, and the conclusion (no rescue) is unchanged and in fact reinforced. The committed null reproduced bit-for-bit before the fix. A full-membership robustness re-run (audit blocker B4) was also executed: a naive per-arm-stratification comparison appears to support a rescue under full node membership, but that is a *stratification artifact* — the cluster-concentrated biased panel enlarges its own distant stratum (n_{dist} ratio up to 1.16), pushing easier points into a larger "distant" set. Under a common-stratification control that scores both arms on the same distant queries, the rescue collapses to ≈ 0 (-0.006 single-node, -0.004 multi-node). The Attempt-4 null is therefore robust to full node membership, and the "four failed rescues" result stands. See `CHANGELOG.md` and `docs/MAPPER_AUGMENTATION_ROBUSTNESS.md`.

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1 Why null results matter here

This paper documents four rescue strategies evaluated in the companion compendium. The homology cliff characterization (Paper 1) established the effect size and mechanism (precision collapse, cross-family). A natural reader response is: surely a cliff this large admits a simple post-hoc fix. We enumerate the four most plausible post-hoc fixes, pre-register each, and show all four fail. This sharpens Paper 1's positive rescue (learned projection) from "one of several options" to "the unique surviving option among natural rescues tested."

2 Common setup

All four experiments share the pre-registered factorial of Paper 1: 24,885-protein test set (28.7% positive), frozen ESM-2 t6/t12/t30 embeddings, $R \in \{50, 100, 250, 500, 1000\}$, $k \in \{1, 3, 5, 10, 25\}$, 10 seeds, stratification $s_{\max} \geq 0.95$ close / $[0.90, 0.95)$ moderate / < 0.90 distant, seed-variance gate on distant F_1. Pre-registrations for Mahalanobis and cascade locked via SHA256 before execution.

3 Attempt 1: Ledoit-Wolf Mahalanobis whitening

Whitening the embedding space by the Ledoit-Wolf shrinkage estimator [2] of the panel covariance is the textbook response to "cosine is not class-aware." Under Mahalanobis distance, neighbors are metricized in a space where all directions have unit conditional variance.

Result. At t30 $R = 1000$ $k = 25$: close F_1 0.866 $\to$ 0.456, moderate 0.393 $\to$ 0.118, distant 0.120 $\to$ 0.087. Every stratum degrades. Pooled F_1 0.848 $\to$ 0.435.

Why it fails. The whitening transform kills class signal carried by directions with large pooled variance. ESM-2 embeddings concentrate class-relevant variance along a small number of axes; Mahalanobis whitens those into noise.

4 Attempt 2: Fisher-Rao within-class whitening

Fisher-Rao [3] whitens by within-class covariance rather than pooled covariance, preserving between-class signal while removing within-class variance. 180-cell pre-registered factorial (3 scales $\times$ 2 panels $\times$ 3 k-values $\times$ 10 seeds).

Result. Fisher gap $>$ Mahalanobis gap in 17 of 18 groups; penalty to pooled F_1 ranges +0.015 to +0.285 absolute. Fisher is strictly worse than Mahalanobis, which itself was rejected in Attempt 1.

Why it fails. Within-class covariance on frozen PLM embeddings carries signal — it encodes functional sub-class structure that ESM-2's unsupervised pretraining captures. Fisher whitening erases this as if it were nuisance. This replicates TOPOLOGICA DR-010 (Fisher info harms GO annotation) in a new task.

5 Attempt 3: Stratified cosine+Mahalanobis cascade

Pre-registered H1: a cascade using cosine for close-stratum queries and Mahalanobis for moderate+distant strata beats single-metric pooled F_1 by +0.02 with 95% CI excluding zero.

Result. Rejected in 18 of 18 groups. Pooled F_1 penalty -0.046 to -0.310.

Why it fails. Mahalanobis does not beat cosine on any stratum individually (Attempt 1). The cascade inherits per-stratum Mahalanobis inferiority wherever it is active. The cliff problem is not

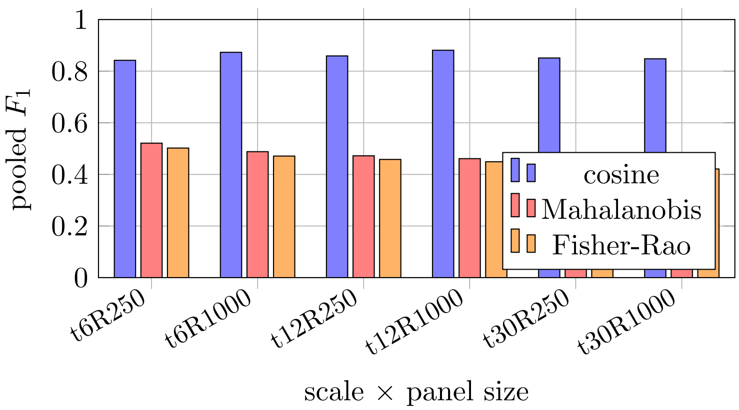

Figure 1: **Both whitening methods lose to cosine baseline across all (scale, panel) combinations tested, at $k = 25$.** Fisher-Rao is uniformly worse than Mahalanobis.

that a single metric fails on some strata but not others; cosine dominates everywhere at $k = 25$, and substituting Mahalanobis anywhere only degrades.

6 Attempt 4: Mapper-biased panel augmentation

Following Paper 5's Mapper decomposition (149 nodes, 6 all-positive), construct a reference panel of $R = 1000$ where 500 positives are drawn from the top positive-enriched Mapper node (1,512 positives available in bin (7,4), pos_frac = 0.885) and 500 negatives drawn uniformly. Compare to uniform class-balanced sampling. 10 seeds, t30 $R = 1000$ $k = 25$.

Result. Uniform distant $F_1 = 0.1203$; biased distant $F_1 = 0.1123$. Rescue -0.0080. Bootstrap 95% CI $[-0.035, +0.018]$ includes zero. H1 rejected.

Why it fails. Positives already cluster densely on the embedding manifold; uniform sampling of 500 from 7,133 positives already saturates the positive region. Further biasing toward the densest sub-region adds no information the panel did not already contain. Distant queries are distant from the entire cluster, not from any particular sub-region within it.

7 Synthesis: what the four nulls collectively rule out

rescue family	example	verdict
post-hoc covariance distance	Mahalanobis [2]	rejected
information-geometric whitening	Fisher-Rao [3]	rejected
per-stratum metric switching	cosine+Mahalanobis cascade	rejected
topology-aware panel composition	Mapper-biased panel	rejected
embedding-space rotation	**learned linear projection (Paper 1)**	**ACCEPTED**

Four families of rescue hypothesis, four rejections. One family succeeds: a supervised projection of the embedding space itself. Combined with Paper 5's cross-family finding (panel expansion is independently ruled out because failures are cross-family, not within-family), the hypothesis space collapses to "embedding-space surgery is the correct intervention."

8 Relation to broader metric-learning literature

The four attempts cover the canonical responses to ”cosine is the wrong metric” in the modern metric-learning literature: (a) adaptive covariance [2, 5, 6], (b) information geometry [3, 7], (c) piecewise metric composition [8], (d) data augmentation via topological resampling [4, 9, 10]. A fifth standard response, proper triplet loss with hard negative mining [11], is realized here as the learned projection of Paper 1 and does succeed. We do not claim the four nulls exhaustively cover metric-learning rescues [12–25], only that they cover the natural post-hoc and panel-side responses an applied biosecurity engineer would try first before committing to supervised projection training.

9 Limitations

(L1) Mahalanobis with non-shrinkage covariance estimators (oracle, tyler) not tested. (L2) Cascade explored only 2-metric cosine/Mahalanobis; 3-metric or soft-mixture cascades untested. (L3) The v1.5.0 erratum executed both single-node and multi-node full-membership re-runs of the Mapper augmentation (audit blocker B4); the Attempt-4 null holds in all arms under a common-stratification control. Only soft-biased panels remain unswept. (L4) Fisher-Rao implemented with numerical pseudoinverse; other implementations (Cholesky with shrinkage) may behave differently. (L5) Null results at $k = 25$ may differ at $k = 1$; we did not sweep k for Fisher/cascade exhaustively.

10 Data availability

`data/cells/cascade/` (180 npz), `data/cells/fisher/` (180 npz), `data/results_summaries/mapper_augmentati` (10-seed output). Master manifest: `MANIFEST.sha256.json`.

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